The documentation and process conversion measures necessary to comply with this document shall be completed by 22 November 2025.

INCH-POUND

MIL-PRF-19500/116U
22 August 2025
SUPERSEDING
MIL-PRF-19500/116T
w/AMENDMENT 4
16 December 2023

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, SILICON, SWITCHING, TYPES 1N4148, 1N914, AND 1N4531
THROUGH HOLE, SURFACE MOUNT AND UNENCAPSULATED DIE,
QUALITY LEVELS JAN, JANTX, JANTXV, JANHC, AND JANKC

Device type JANS1N4148-1 and JANS1N4148UR-1 are inactive for new design (see 6.6).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon switching diodes. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type.

1.2 Physical dimensions. The device package for the encapsulated device type are as follows: Axial leaded DO-35, DO-34 in accordance with figure 1, DO-213AA in accordance with figure 2, UB in accordance with figure 3, UB2 in accordance with figure 4, UBC in accordance with figure 5, and UD in accordance with figure 6. The dimensions and topography for JANHC and JANKC un-encapsulated die are as follows: A and B version die in accordance with figure 7, C and D version die in accordance with figure 8, E version die in accordance with figure 9, and F version die in accordance with figure 10.

1.3 Maximum ratings. Unless otherwise specified, at T_A = +25°C.

Type	V_{BR}	V_{RWM}	$I_{O(PCB)}$ T_A = 75°C (1)	I_{FSM} t_P = 8.3ms	T_{STG} & T_J	$R_{\theta JA(PCB)}$ (2) (3) (4)	$R_{\theta JL}$ L = .375 inch (9.53 mm) (2)	$R_{\theta JEC}$ (UR) (2)	$R_{\theta JSP}$ (2) (4)
	V dc	V (pk)	mA	A (pk)	°C	°C/W	°C/W	°C/W	°C/W
1N4148-1	100	75	200	2	-65 to +175	325	250		
1N4148UR-1								100	
1N4148UD									80
1N4148UB, 1N4148UB2, 1N4148UB2R, 1N4148UBCA, 1N4148UBCC, 1N4148UBD, 1N4148UBCCC, 1N4148UBCD					-65 to +200	325			120

See footnotes at end of table.

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FSC 5961

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1.3 Maximum ratings. Unless otherwise specified T_A = +25°C – continued.

Type	V_{BR}	V_{RWM}	$I_{O(PCB)}$ T_A = 75°C (1)	I_{FSM} t_P = 8.3ms	T_{STG} & T_J	$R_{\theta JA(PCB)}$ (2) (3) (4)	$R_{\theta JL}$ L = .375 inch (9.53 mm) (2)	$R_{\theta JEC}$ (UR) (2)	$R_{\theta JSP}$ (2) (4)
	V dc	V (pk)	mA	A (pk)	°C	°C/W	°C/W	°C/W	°C/W
1N4531	100	75	200	2	-65 to +175	325	250		
1N4531UR								100	
1N4531UD						325			80
1N914						325	250		
1N914UR								100	
1N914UD						325			80

(1) For temperature-current derating curves, see figures 11 and 12.
(2) See figures 13, 14, 15 and 16 for thermal impedance curves.
(3) T_A = +75°C for both axial leaded and metal electrode leadless face (MELF) UR diodes on printed circuit board (PCB), PCB = FR4, .0625 inch (1.59 mm) 1-layer 1-Oz Cu, horizontal, in still air; pads for UR = .061 inch (1.55 mm) x .105 inch (2.67 mm); pads for axial = .092 inch (2.34 mm) diameter, strip = .030 inch (0.76 mm) x 1 inch (25.4 mm) long, lead length L ≤ 0.187 inch (≤ 4.75 mm); $R_{\theta JA}$ with a defined PCB thermal resistance condition included, is measured at I_O = 200 mA dc.
(4) $R_{\theta JSP}$ refers to thermal resistance from junction to the solder pads of the UB, UB2, and UD package.

1.4 Primary electrical characteristics. Unless otherwise indicated, at T_A = +25°C.

Type (1)	V_{F1}		V_{F2}		I_{R1} at V_R = 20 V dc	I_{R2} at V_R = 75 V dc
	I_F mA dc	V dc	I_F mA dc	V dc	nA dc	nA dc
1N4148-1	10	0.8	100	1.2	25	500
1N914	10	0.8	50	1.2	25	500
1N4531	10	0.8	100	1.2	25	500

Type (1)	I_{R3} at V_R = 20 V dc T_A = 150°C	I_{R4} at V_R = 75 V dc T_A = 150°C	t_{fr} at V_{fr} = 5.0 V dc (pk) and I_F = 50 mA dc	t_{rr}
	µA dc	µA dc	ns	ns
1N4148-1	35	75	20	5
1N914	35	75	20	5
1N4531	35	75	20	5

(1) Primary electrical characteristics for surface mount devices are equivalent to the corresponding non-surface mount devices unless otherwise noted.

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level designators.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", and "JANTXV".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the device types of semiconductor covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the semiconductor covered by this specification sheet are as follows: "914", "4148", and "4531".

1.5.3 Suffix symbols. The following suffix letters are incorporated in the PIN for this specification sheet.

1.5.3.1 Package designator for round body devices. The suffix symbols (or lack thereof) that designate the package outline for the round body devices covered by this specification sheet are as follows.

	A blank first suffix symbol indicates an axial through-hole mount DO-204AG or DO-204AH package (see figure 1).
-1	A dash-one (-1) designator identifies the package as a round body package that uses a metallurgically bonded double plug, category 2 or 3 construction (see figure 1).
UR	A UR designator identifies the package as a surface mount DO 213AA (see figure 2).

1.5.3.2 Package designator for bottom pad surface mount packages (type 1N4148 only). The suffix symbols used to identify the package outline for the bottom pad surface mount devices covered by this specification sheet are as follows:

UB	Indicates a 3 terminal, 4 pad surface mount package (see figure 3).
UB2	Indicates a 2 terminal, 4 pad surface mount package (see figure 4).
UBC	Indicates a 3 terminal, 4 pad surface mount package with ceramic lid (see figure 5).
UD	Indicates a 2 terminal, 3 pad surface mount package (see figure 6).

1.5.3.3 Internal circuit schematic designators (1N4148 with UB, UB2 and UBC only). The suffix symbols that designate the internal circuit schematic of the bottom pad surface mount packages are as follows:

	Indicates a unidirectional, single diode circuit (UB and UB2 packages only) (see figures 3 and 4).
R	Indicates a unidirectional, reverse polarity, single diode circuit (UB2 package only)(see figure 4).
CA	Indicates dual diode with a common anode pin (UB package only) (see figure 3).
CC	Indicates a dual diode with a common cathode pin (UB and UBC only) (see figures 3 and 5).
D	Indicates a dual diode doubler (UB and UBC packages only) (see figures 3 and 5).

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1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A", "B", "C", "D", and "E".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

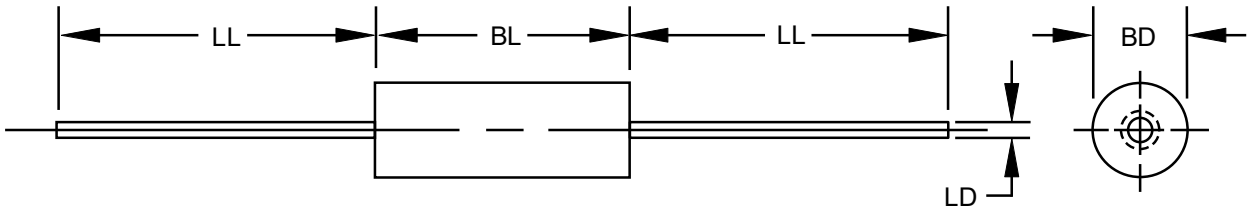
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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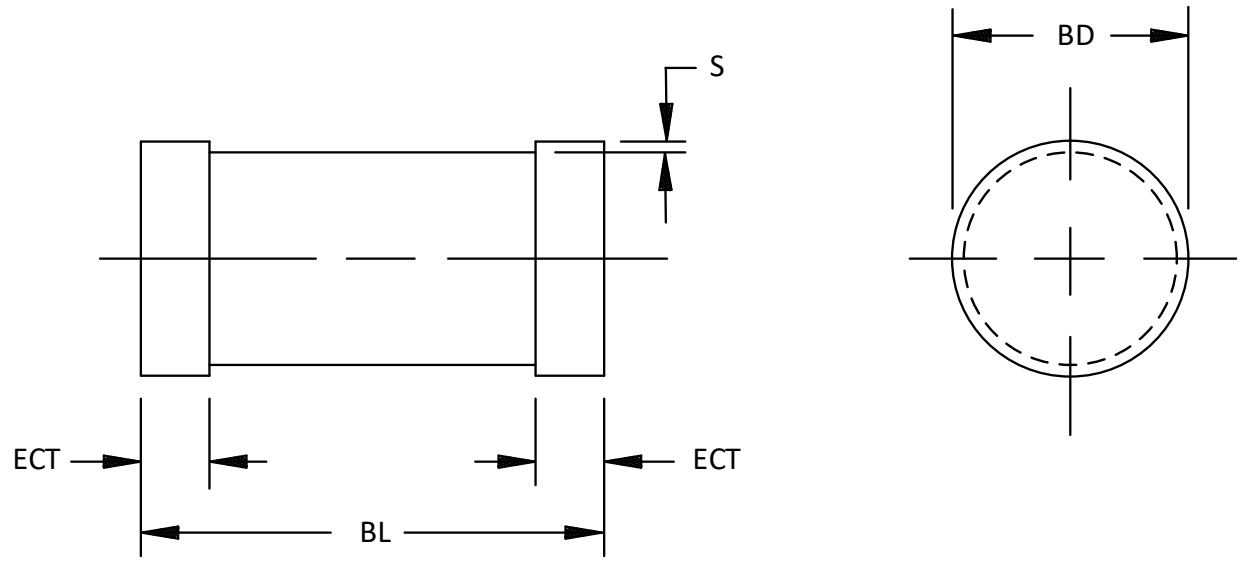
TYPES 1N4148-1, 1N914, AND 1N4531.

Types	Dimensions				
	Symbols	inch Min	inch Max	mm Min	mm Max
	BD	.056	.075	1.42	1.91
1N4148-1	BL	.140	.180	3.56	4.57
1N914	LD	.018	.022	0.46	0.56
(DO-35)	LL	1.000	1.500	25.40	38.10
	BD	.050	.075	1.27	1.90
1N4531	BL	.080	.120	2.03	3.05
(DO-34)	LD	.018	.022	0.46	0.56
	LL	1.000	1.500	25.40	38.10

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension BD shall be measured at the largest diameter.
4. Package contour optional within dimension BD and length dimension BL.

FIGURE 1. Physical dimensions for DO-34 and DO-35 axial leaded packages.

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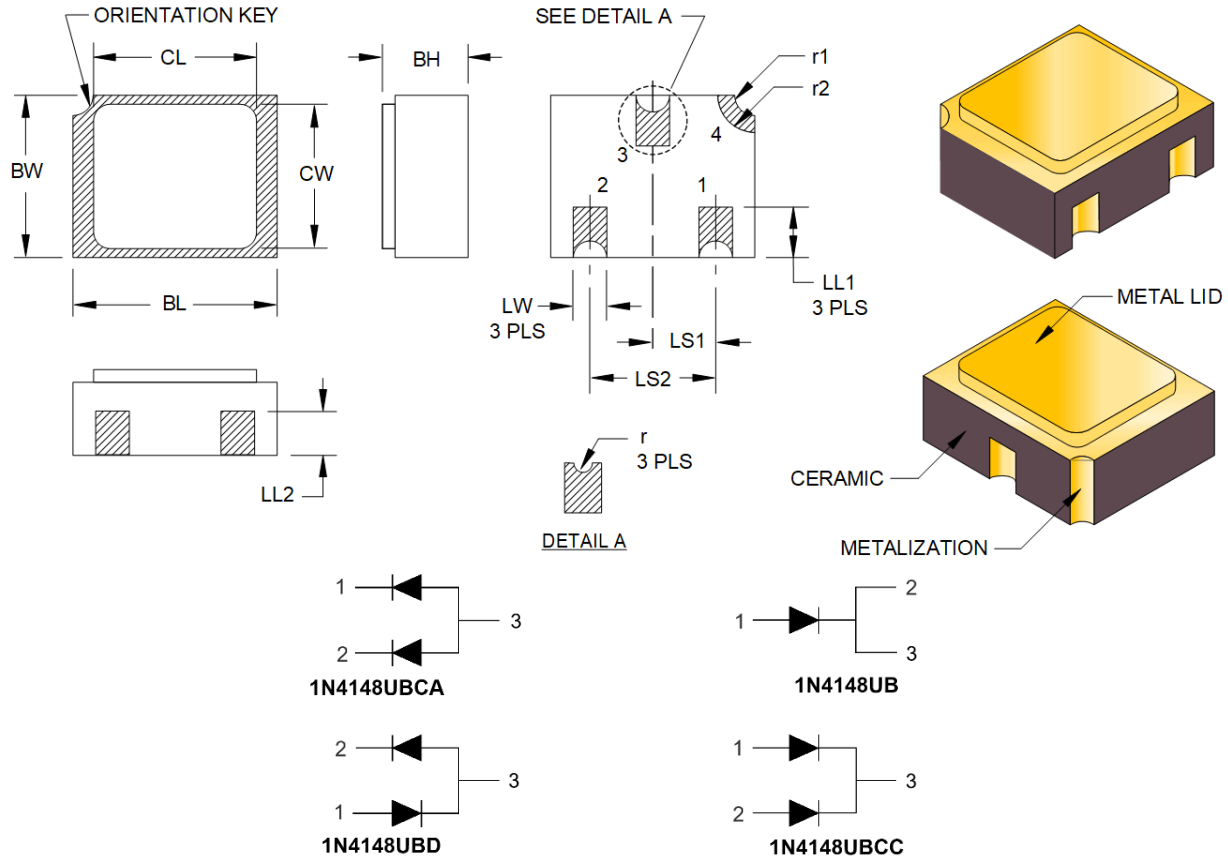
TYPES 1N914UR, 1N4148UR-1, AND 1N4531UR.

Symbol	Dimensions				
	inch	inch	mm	mm	Note
	Min	Max	Min	Max	
BD	.063	.067	1.60	1.70	
BL	.130	.146	3.30	3.71	
ECT	.016	.022	0.41	0.56	
S	0		0		3

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension S is optional however the glass body diameter shall not exceed endcap diameter.

FIGURE 2. Physical dimensions of round end cap, UR (DO-213AA) surface mount package.

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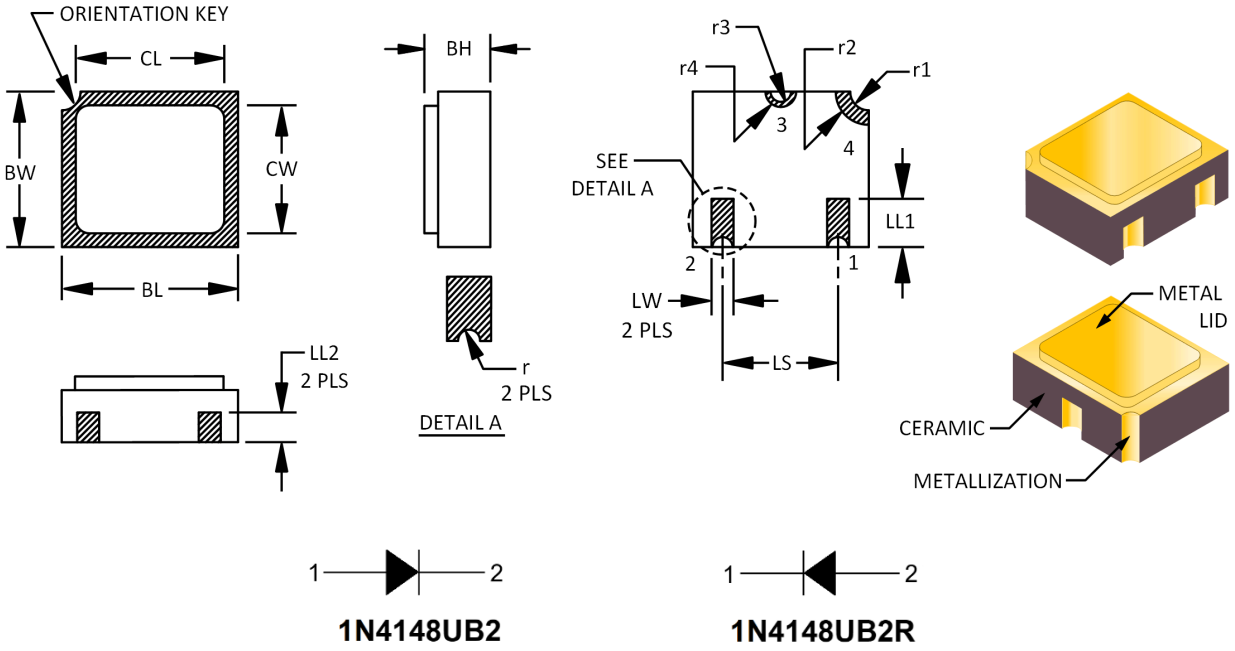


Symbol	Dimensions				Symbol	Dimensions			
	inch	inch	mm	mm		inch	inch	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.040	.056	1.02	1.42	LS1	.035	.040	0.89	1.02
BL	.115	.128	2.92	3.25	LS2	.071	.079	1.80	2.01
BW	.085	.108	2.16	2.74	LW	.016	.024	0.41	0.61
CL		.128		3.25	r		.008		0.20
CW		.108		2.74	r1		.012		0.31
LL1	.022	.038	0.56	0.97	r2		.022		0.56
LL2	.017	.035	0.43	0.89					

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Metal lid, ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = shielding, connected to the lid.

FIGURE 3. Physical dimensions, UB metal lid surface mount packages.

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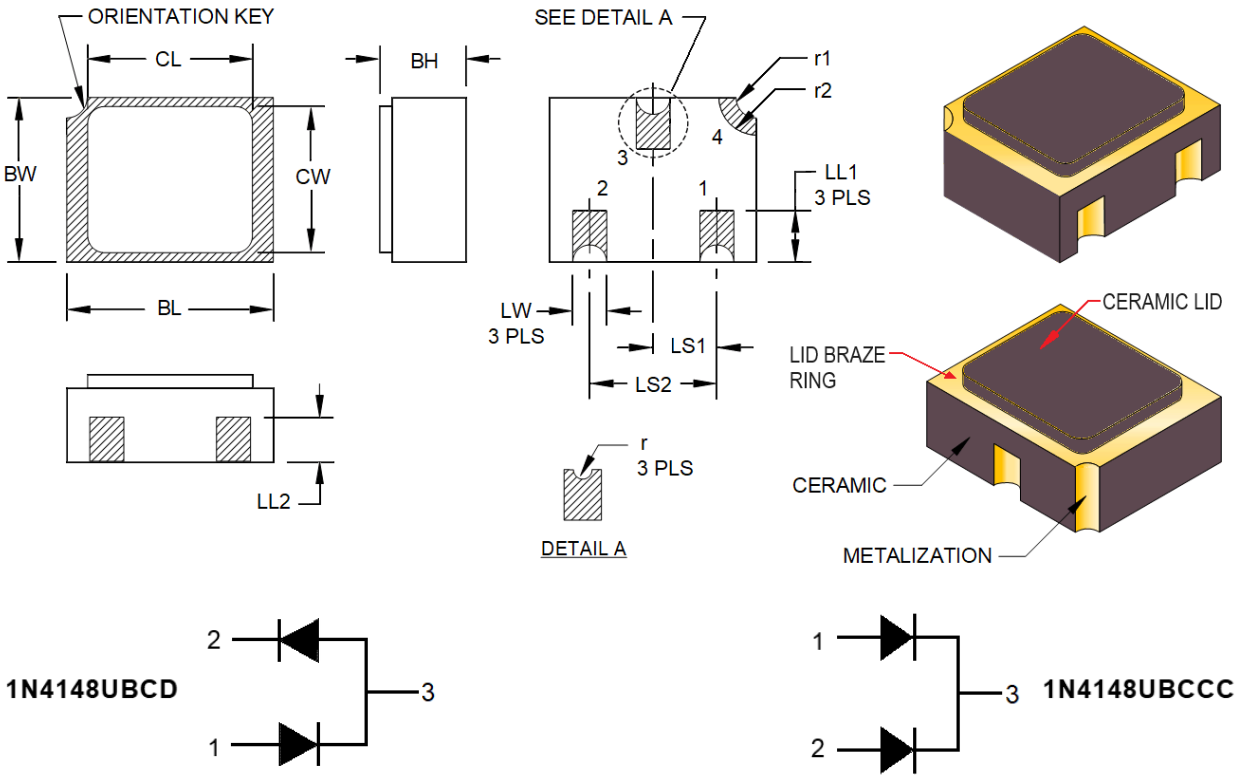


Symbol	Dimensions				Symbol	Dimensions			
	inch	inch	mm	mm		inch	inch	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.046	.056	1.17	1.42	LS	.071	.079	1.80	2.01
BL	.115	.128	2.92	3.25	LW	.016	.024	0.41	0.61
BW	.085	.108	2.16	2.74	r	.008	TYP	0.20	TYP
CL		.128		3.25	r1	.012	TYP	0.31	TYP
CW		.108		2.74	r2	.022	TYP	0.56	TYP
LL1	.022	.038	0.56	0.96	r3	.008	TYP	0.20	TYP
LL2	.017	.035	0.43	0.89	r4	.012	TYP	0.31	TYP

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Metal lid, ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = shielding, connected to the lid.

FIGURE 4. Physical dimensions, 2 terminal UB2 surface mount package.

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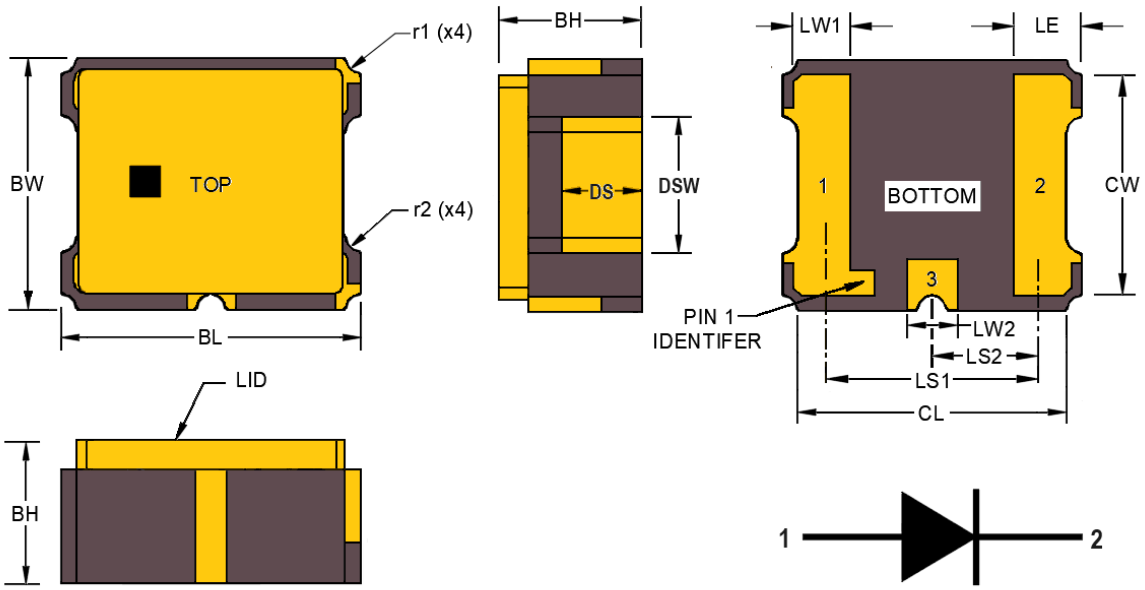
Symbol	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
BH	.046	.071	1.17	1.80
BL	.115	.128	2.92	3.25
BW	.085	.108	2.16	2.74
CL		.128		3.25
CW		.108		2.74
LL1	.022	.038	0.56	0.96
LL2	.017	.035	0.43	0.89

Symbol	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
LS_1	.036	.040	0.91	1.02
LS_2	.071	.079	1.81	2.01
LW	.016	.024	0.41	0.61
r		.008		.203
r1		.012		.305
r2		.022		.559

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic lid, ceramic package only.
3. Hatched areas on package denote metallized areas. Pin 4 is connected to lid braze ring.

FIGURE 5. Physical dimensions, UBC ceramic lid surface mount.

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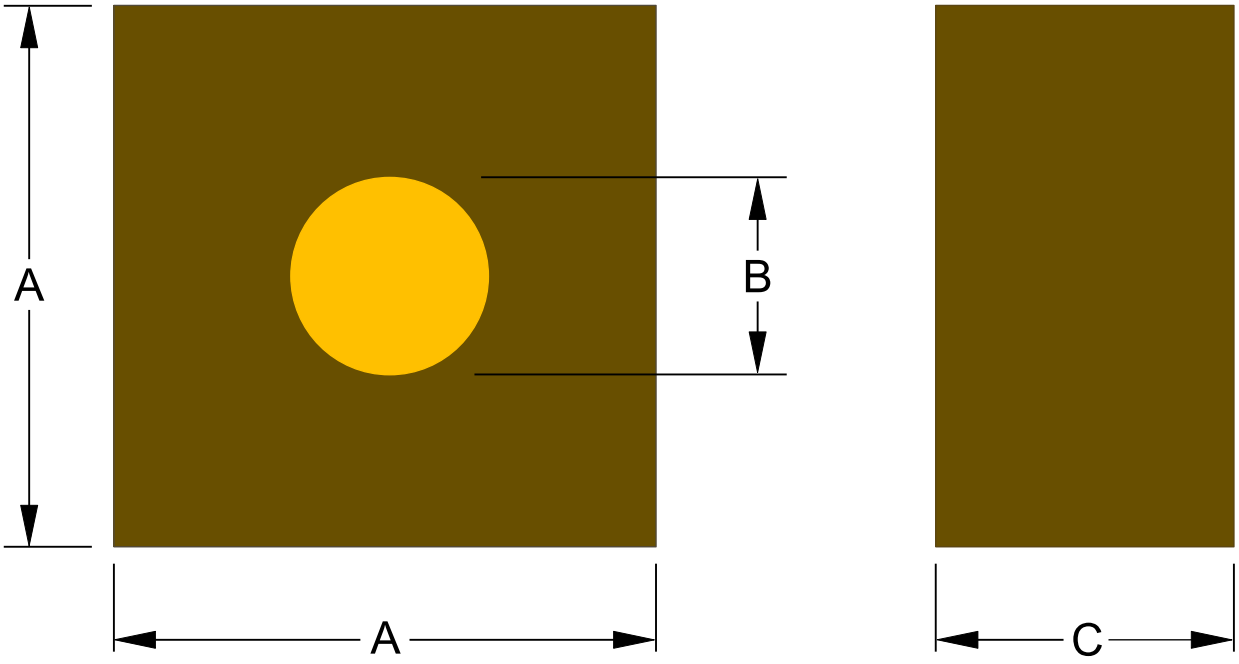


Symbol	inch	inch	mm	mm	Symbol	inch	inch	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	0.048	0.062	1.186	1.590	LE	0.021	0.033	0.540	0.840
BL	0.112	0.128	2.850	3.250	LS1	0.085	0.093	2.160	2.360
BW	0.092	0.108	2.340	2.740	LS2	0.043	0.047	1.080	1.180
CL	0.103	0.113	2.620	2.880	LW1	0.019	0.027	0.490	0.690
CW	0.083	0.093	2.120	2.360	LW2	0.019	0.021	0.490	0.530
DS	0.028	0.034	0.720	0.880	r1		0.006		0.150
DSW	0.051	0.057	1.330	1.430	r2		0.006		0.150

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Pad 3 is connected to the lid braze ring for shielding.
3. Cathode marked with a square.
4. Pads 1 and 2 layout equivalent to 1N914UR, 1N4148UR-1, and 1N4531UR.
5. Diode outline shown in ISO third angle projection.

FIGURE 6. Physical dimensions of UD surface mount package.

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Dimensions for JANHCA and JANKCA die

Letter	inch	inch	mm	mm
	Min	Max	Min	Max
A	.0130	.0170	.330	.432
B	.0059	.0061	.150	.155
C	.008	.012	0.20	0.30

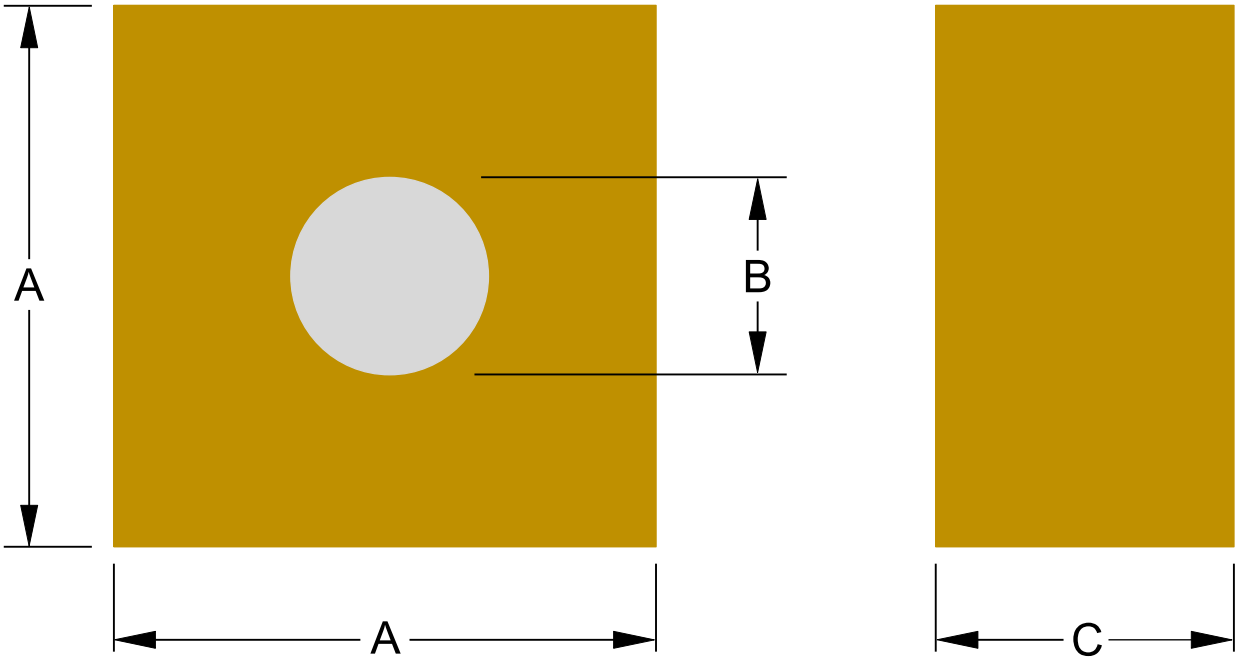
Dimensions for JANHCB and JANKCB die

Letter	inch	inch	mm	mm
	Min	Max	Min	Max
A	.014	.018	.360	.460
B	.005	.007	.120	.180
C	.008	.012	0.20	0.30

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Element evaluation accomplished utilizing TO-5 package.
3. The physical characteristics of the die are:
   Metallization:
   Top (anode): Al
   Back (cathode): Au
   Al thickness: 25,000 Å minimum.
   Gold thickness: 4,000 Å minimum.

FIGURE 7. Physical dimensions for JANHCA, JANKCA, JANHCB and JANKCB die.

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Dimensions for JANHCC and JANKCC die

Letter	inch	inch	mm	mm
	Min	Max	Min	Max
A	.017	.021	.432	.533
B	.008	.010	.203	.254
C	.007	.011	0.178	0.279

Dimensions for JANHCD and JANKCD die

Letter	inch	inch	mm	mm
	Min	Max	Min	Max
A	.017	.021	.432	.533
B	.008	.010	.203	.254
C	.007	.011	0.178	0.279

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Element evaluation accomplished utilizing TO-39 package.
3. The metallization characteristics of the die are as follows:

Metallization for JANHCC and JANKCC die

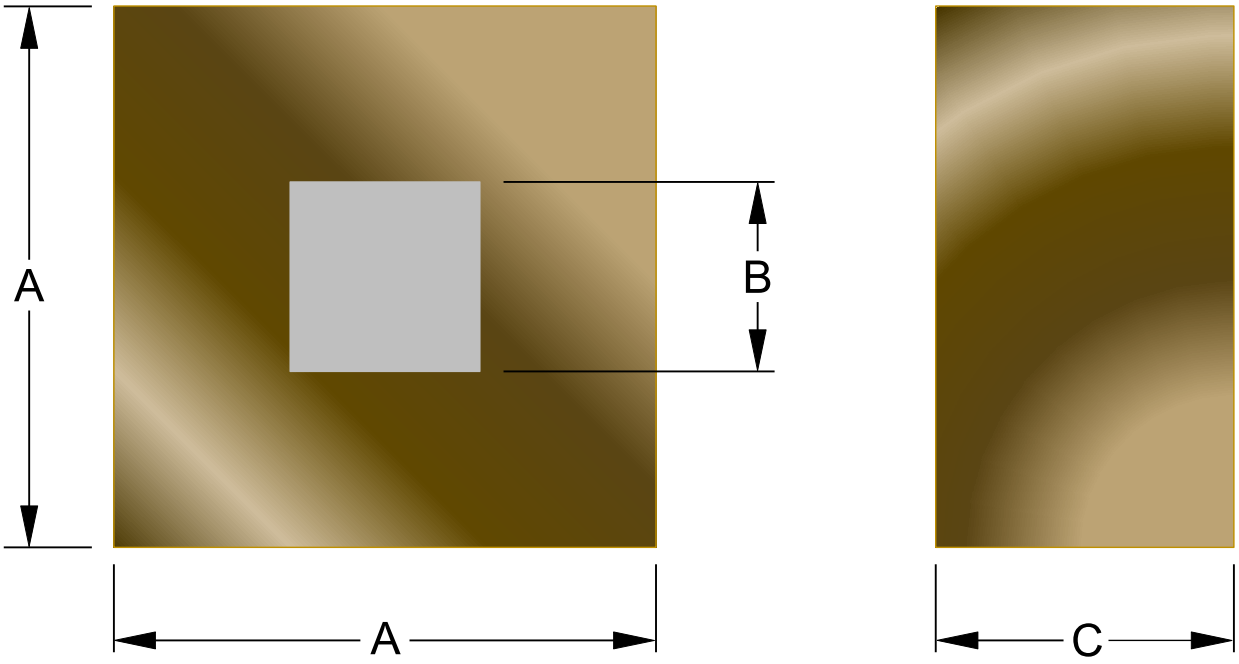
Top (anode): Al
Al thickness: 34,000 Å minimum.
Back (cathode): Ti / Ni / Au
Ti / Ni / Au thickness (1,200 Å / 1,800 Å / 4,000 Å) nominal.

Metallization for JANHCD and JANKCD die

Top (anode): Al
Al thickness: 34,000 Å minimum.
Back (cathode): Ti / Au
Ti / Au thickness: Ti / Au (200 Å / 4,350 Å) nominal.

FIGURE 8. Physical dimensions for JANHCC, JANKCC, JANHCD and JANKCD die.

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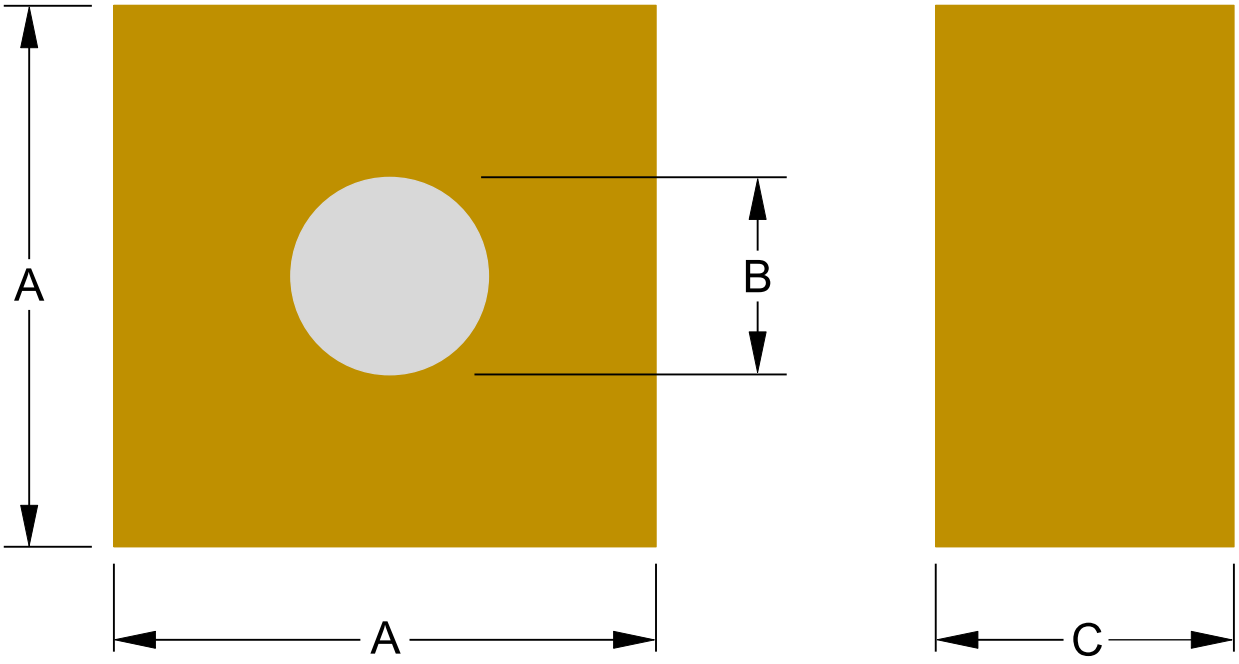
Letter	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.015	.019	.381	.483
B	.004	.006	.102	.152
C	.006	.010	.152	.254

NOTES

1. Dimensions are in inches. Millimeters are given for general information only.
2. Element evaluation accomplished utilizing any package available.
3. The metallization characteristics of the die are as follows:
   - Top metal (anode): Al-Si
   - Back metal (cathode): Au
   - Al-Si thickness: 14,400 Å minimum.
   - Au thickness: 5,800 Å minimum.

FIGURE 9. Physical dimensions for JANHCE and JANKCE (E-version) die.

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Letter	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.013	.017	.330	.432
B	.007	.009	.178	.229
C	.007	.011	.178	.279

NOTES
1. Dimensions are in inches. Millimeters are given for general information only.
2. Element evaluation accomplished utilizing a TO-39 package.
3. The metallization characteristics of the die are as follows:
   - Top metal (anode): Ali
   - Back metal (cathode): Ti / Au
   - Al thickness: 34,000 Å minimum.
   - Ti / Au thickness: 200 Å / 4,350 nominal.

FIGURE 10. Physical dimensions for JANHCF and JANKCF (F-version) die.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

SP Solder pad on UB, UB2, UBC, and UD devices.
V_{fr} Forward recovery voltage. Specified maximum forward voltage used to determine forward recovery time.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (axial leaded), figure 2 (UR, DO-213AA), figure 3 (UB), figure 4 (UB2), figure 5 (UBC), figure 6 (UD) and figures 7 through 10 for un-encapsulated die.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction.

3.4.2.1 Axial leaded and round body surface mount. Axial leaded versions and round body surface mount shall be metallurgically bonded, double plug construction, in accordance with the requirements of MIL-PRF-19500. All glass diodes shall be designed with sufficient thermal compensation in the axial direction to optimize tensile and compressive stresses. Dimensional analysis is required of all materials used to achieve axial thermal compensation. Dimensional tolerances and corresponding coefficient of thermal expansion (CTE) shall be documented on the DLA Land and Maritime Design and Construction Form 36D and shall be approved by the qualifying activity to maintain qualification. Dimensional tolerances shall be sufficiently tight enough to prevent excessive stresses due to the inherent CTE mismatch. The UR version shall be structurally identical to the axial leaded versions except for end-cap lead attachment.

3.4.2.2 Bottom solder pad surface mount. The UB, UB2, UBC, and UD version devices die shall be eutectically mounted and wire bonded in a ceramic package.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. Initial container package marking shall be in accordance with MIL-PRF-19500. The prefixes JAN, JANTX, or JANTXV may be abbreviated as J, JX, or JV, respectively. (Example: The PIN for an axial leaded device may be reduced to from JANTXV1N4148-1 to JV4148). All marking which is omitted from the body of the device, except for polarity and serial numbers, shall appear on the initial container.

3.5.1 Axial leaded devices. For axial leaded versions only, the manufacturer's identification and date code shall be marked on the devices. The polarity shall be indicated with a contrasting color band to denote the cathode end. No color coding shall be permitted for part identification.

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3.5.2 UR devices. For 'UR' version devices only, all marking, except polarity, may be omitted from the body, but shall be retained on the initial container. Polarity marking of 'UR' devices shall consist as a minimum, a band or three contrasting dots around the periphery of the cathode. At the option of the manufacturer, 'UR' devices may include laser marking on an end-cap, to include PIN and lot date code for all levels.

3.5.3 UB devices. The 'UB', 'UB2' and 'UBC' devices do not require polarity marking.

3.5.4 UD devices. The 'UD' devices shall mark the cathode with a square symbol on the lid.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I. Electrical characteristics for surface mount versions are identical to the corresponding axial leaded versions unless otherwise specified.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation of un-encapsulated die (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not require the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with appendix G of MIL-PRF-19500 and as specified herein.

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4.3 Screening (quality levels JANTXV, JANTX, and JAN only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screening	Measurements for JANTXV and JANTX level
3c (1)	Thermal impedance (see 4.3.1)
9	Not required
10	Method 1038 of MIL-STD-750, condition A
11 (2)	I_{R1} and V_{F1}
12	Method 1038 of MIL-STD-750, condition B, see 4.3.2
13 (3) (4)	Subgroup 2 of table I herein; ΔI_{R1} ≤ 100 percent of initial value or 15 nA dc, whichever is greater; ΔV_{F1} ≤ 25 mV dc

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) Test within 24 hours after removal from test.
(3) When thermal impedance is performed prior to screen 13, it is not required to be repeated in screen 13.
(4) PDA ≤ 5 percent.

4.3.1 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, as applicable, using the guidelines in that method for determining I_H and I_M. t_{MD} shall be 70 µs maximum, t_H shall be 10 ms maximum. See table E-IX of MIL-PRF-19500 and subgroup 4 of table II herein.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.3): V_R = rated V_{RWM}; f = 50 to 60 Hz; I_O = 200 mA dc or I_F = 200 mA dc minimum. T_A = +75°C maximum. The maximum current density of small die shall be submitted to the qualifying activity for approval. Alternate mounting conditions shall be submitted to the qualifying activity for approval. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, and mounting conditions) may be used. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.3 JAN testing. JAN level product shall have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening level requirements. Electrical testing shall be in accordance with table I, subgroup 2 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500, table I herein, and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Leaded samples from the same lot may be used in lieu of UR suffix sample for life test.

Subgroup	Method	Conditions
B2	1056	0°C to +100°C, 10 cycles.
B2	1051	-55°C to +175°C, 45 cycles, including screening.
B2	2005	I_F = 100 mA, axial tensile stress = 8 lbs (3.63 kg), T_A = +150°C; (Not applicable to UR or UB, UB2, UBC, and UD package).
B3	1027	V(pk) = rated V_{RWM}; f = 50 to 60 Hz; I_O = 200 mA dc minimum; adjust T_A or I_O to obtain a minimum T_J of +150°C. (See 4.5.3.)
B4	2101	Decap analysis; scribe and break (not applicable for UB, UB2, UBC and UD).
B4	2075	For UB, UB2, UBC and UD devices only.
B6	1032	T_A = +175°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Conditions
C2	1056	0°C to +100°C, 10 cycles.
C2	1051	-55°C to +175°C, 45 cycles including screening.
C2	2036	Tension = condition A; weight = 10 pounds (4.54 kg), t = 15 s; Lead fatigue = condition E (not applicable to UR, UB, UB2, UBC and UD suffix types).
C5	4081	L = .375 inch (9.53 mm), $R_{\theta JL}$ = 250°C/W maximum; $R_{\theta JEC}$ = 100°C/W; $R_{\theta JA}$ = 325°C/W; (see 4.3.1), 22 devices, c = 0.
C6	1026	1,000 hours minimum, $V_{(pk)}$ = rated V_{RWM}; f = 50 to 60 Hz; I_O = 200 mA dc minimum; adjust T_A or I_O to obtain a minimum T_J of +150°C (see 4.5.3).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Forward recovery voltage and time. Forward recovery shall be measured as the time interval between zero time and the point where the pulse has decreased to 110 percent of the steady-state value of V_F when I_F = 50 mA dc. The maximum rise time of the response detector shall be 1 ns.

4.5.3 Free air power burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each device under test still sees the full $P_{T(minimum)}$ and that the minimum applied voltage, where applicable, is maintained through-out the burn-in period. Method 3100 of MIL-STD-750 shall be used to measure T_J.

4.5.4 Scope display evaluation. The reverse characteristics shall be inspected and evaluated in accordance with condition A of method 4023 of MIL-STD-750. Each device shall be evaluated only for the following 4 commonly observed faults: drift, unstable (jitter), snap back collapsing $V_{(BR)}$ and floater. Any instance of any of these 4 faults in the trace shall be cause for rejection.

4.6 Element evaluation of un-encapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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TABLE I. Group A inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 4/	3101	See 4.3.1	$Z_{\theta JX}$			°C/W
Forward voltage	4011	Condition B, I_F = 10 mA dc (pulsed, See 4.5.1)	V_{F1}		0.8	V dc
Breakdown voltage	4021	I_R = 100 µA dc	V_{BR1}	100		V dc
Reverse current	4016	Condition A or B; DC method, V_R = 20 V dc	I_{R1}		25	nA dc
Reverse current	4016	Condition A or B, DC method, V_R = 75 V dc	I_{R2}		500	nA dc
Forward voltage	4011	Condition B, pulsed, see 4.5.1				
1N914		I_F = 50 mA dc	V_{F2}		1.2	V dc
1N4148-1, 1N4531		I_F = 100 mA dc	V_{F2}		1.2	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Reverse current	4016	Condition A or B; DC method, V_R = 20 V dc	I_{R3}		35	µA dc
Reverse current	4016	Condition A or B; DC method, V_R = 75 V dc	I_{R4}		75	µA dc
Forward voltage	4011	Condition B, I_F = 10 mA dc (pulsed, see 4.5.1)	V_{F3}		0.8	V dc
Low temperature operation:		T_A = -55°C				
Forward voltage	4011	Condition B, pulsed, see 4.5.1	V_{F4}			
1N914		I_F = 50 mA dc			1.3	V dc
1N4148-1, 1N4531		I_F = 100 mA dc			1.3	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Capacitance	4001	V_R = 0 V dc, f = 1 MHz, V_{SIG} = 50 mVp-p maximum	C_1		4.0	pF
Capacitance	4001	V_R = 1.5 V dc, f = 1 MHz, V_{SIG} = 50 mVp-p maximum	C_2		2.8	pF
Reverse recovery time	4031	Condition A, C ≥ 1 nF, $I_F = I_R$ = 10 mA dc, R_L = 100Ω ±5 percent, $I_{R(REC)}$ = 1.0 mA dc, R ≥ 1,000 Ω	t_{rr}		5	ns
Scope display evaluation	4023	See 4.5.4				
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	Condition A (sine wave), $I_{F(surge)}$ = 2 A(pk), I_O = 0, V_{RM} = 0, 10 surges, 8.3 ms width each, one surge per minute, T_A = +25°C or Condition B (square wave), $I_{F(surge)}$ = 4 A (pk), 10 surges, 1 µs width each, duty factor = 0.0055 percent, T_A = 25°C				
Electrical measurements		See table I, subgroup 2				
Subgroup 7						
Forward recovery voltage and time	4026	I_F = 50 mA dc; (see 4.5.2)	V_{peak} t_{fr}		5.0 20	V (pk) ns

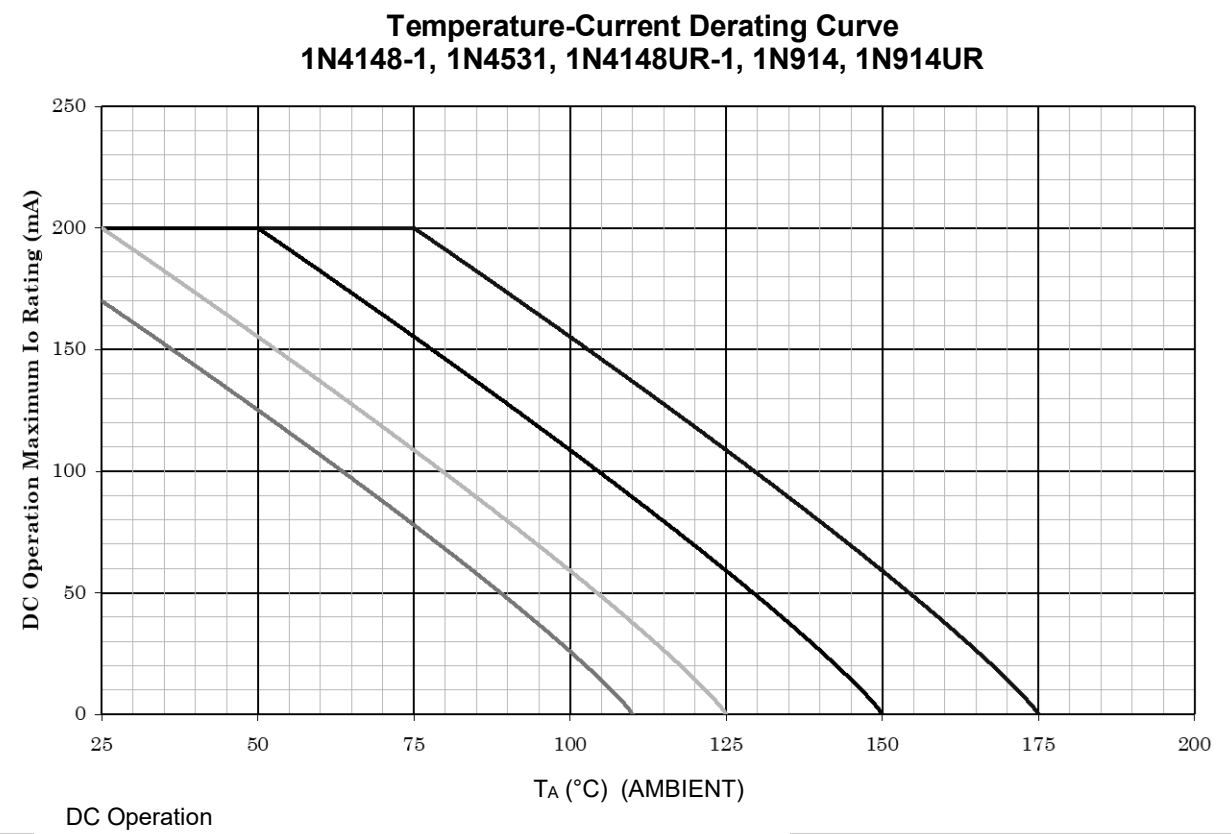
1/ For sampling plan, see MIL-PRF-19500.
2/ UB and UBC suffix devices are to have each diode tested individually.
3/ Electrical characteristics for surface mount versions are identical to the corresponding axial leaded versions unless otherwise specified.
4/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JAN, JANTX, JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			n = 45, c = 0
Thermal shock (glass strain)	1056	100 cycles, 0°C to +100°C	
Temperature cycling	1051	500 cycles, -65°C to +175°C	
Hermetic seal	1071	Gross leak only for axial and UR devices. Fine and gross leak required for UB, UB2, UBC and UD suffix devices.	
Electrical measurement		See table I, subgroup 2	
Subgroup 2			n = 45, c = 0
Intermittent operating life	1037	10,000 cycles; I_f = 300 mA dc, T_{on} = T_{off} = 1 minute.	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			
Thermal resistance	4081	$R_{\theta JSP}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets).	n = 15, c = 0
Thermal impedance curves		See MIL-PRF-19500, table E-IX, group E, subgroup 4.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			
ESD	1020		
Subgroup 8			n = 45
Resistance to glass cracking	1057	Test condition B. Test until failure occurs or to a maximum of 25 cycles, whichever comes first. Not required for UB suffix devices	
Subgroup 9			n = 22, c = 0
Monitored mission temperature cycling	1055	Not required for UB, UB2, UBC and UD suffix devices	
Electrical measurements		See table I, subgroup 2	

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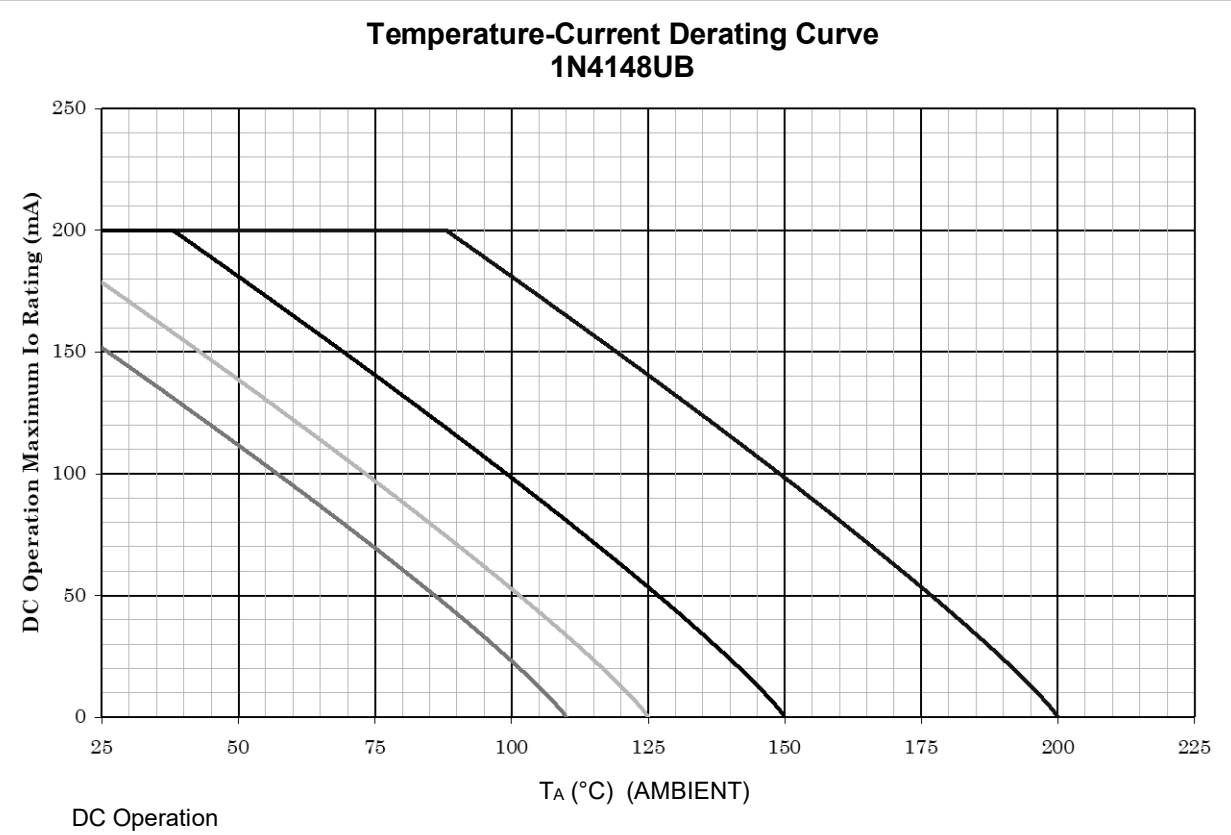


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +175°C) and current rating specified (see 1.3.).
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ +125°C, and 110°C to show current rating where most users want to limit T_J in their application.

FIGURE 11. Temperature-current derating graph for glass round body packages.

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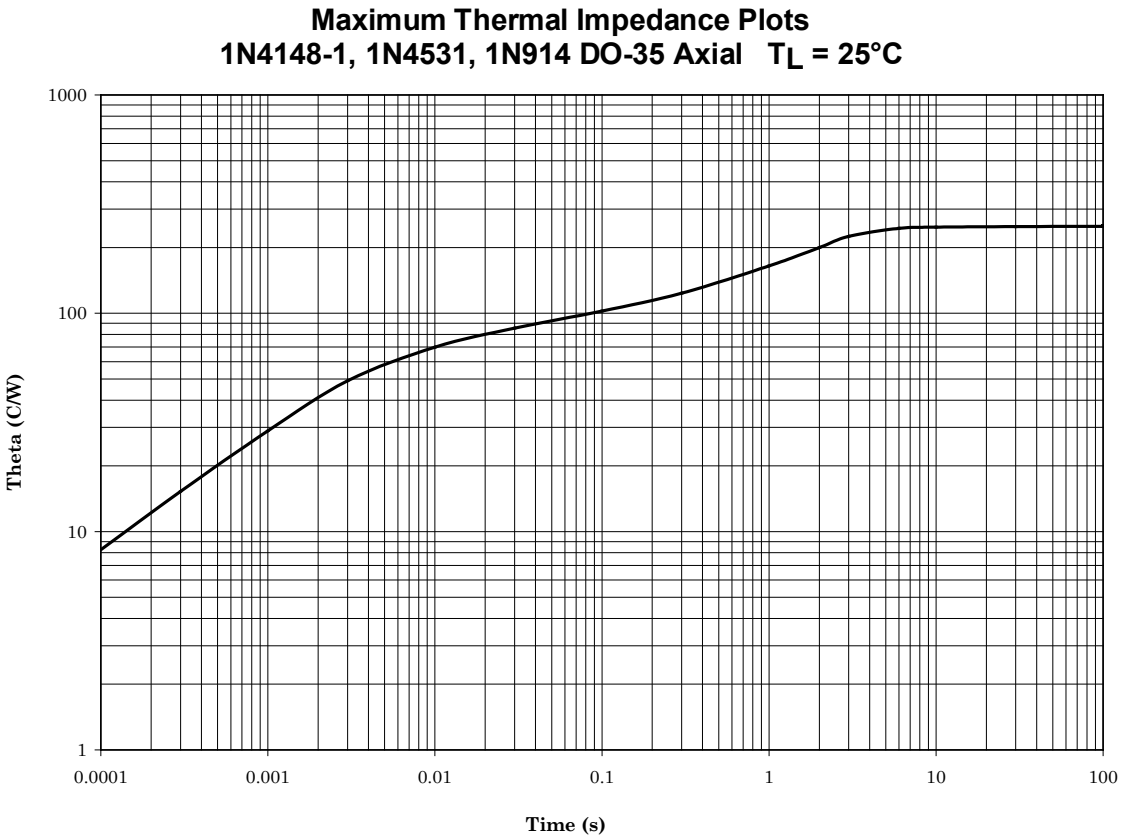


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate current for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +200°C) and current rating specified (see 1.3.).
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ +125°C, and +110°C to show current rating where most users want to limit T_J in their application.

FIGURE 12. Temperature-current derating graph for 1N4148UB surface mount package.

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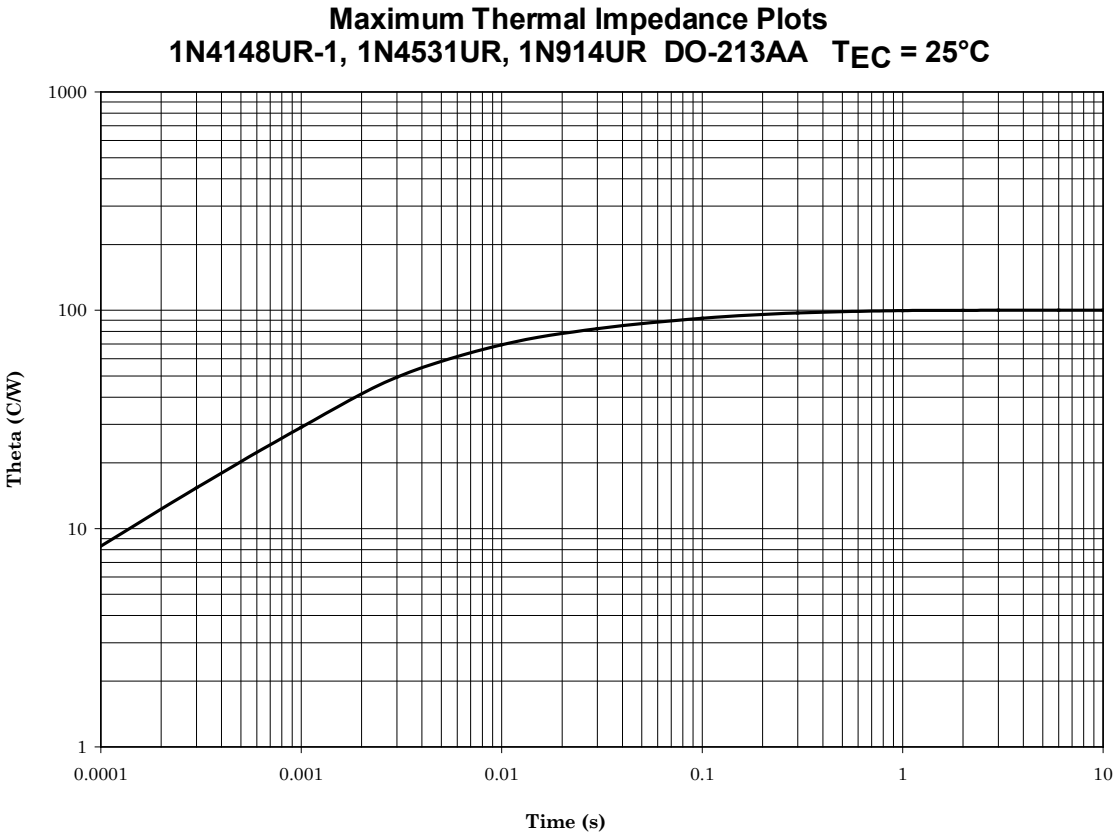


$R_{\theta JL}$ = 250°C/W

NOTE: $Z_{\theta JX}$ = 70°C/W maximum at t_H = 10 ms.

FIGURE 13. Thermal impedance for glass body axial leaded package devices.

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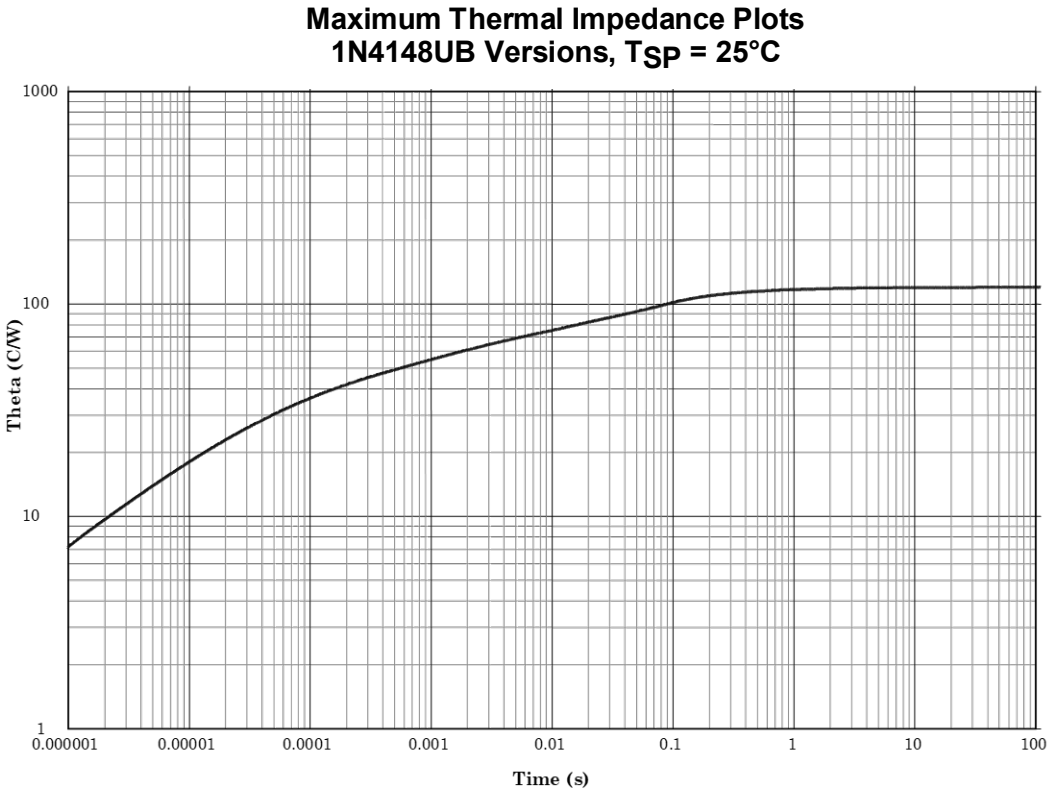


$R_{\theta JEC}$ = 100°C/W

NOTE: $Z_{\theta JX}$ = 70°C/W maximum at t_H = 10 ms.

FIGURE 14. Thermal impedance for UR surface mount packages.

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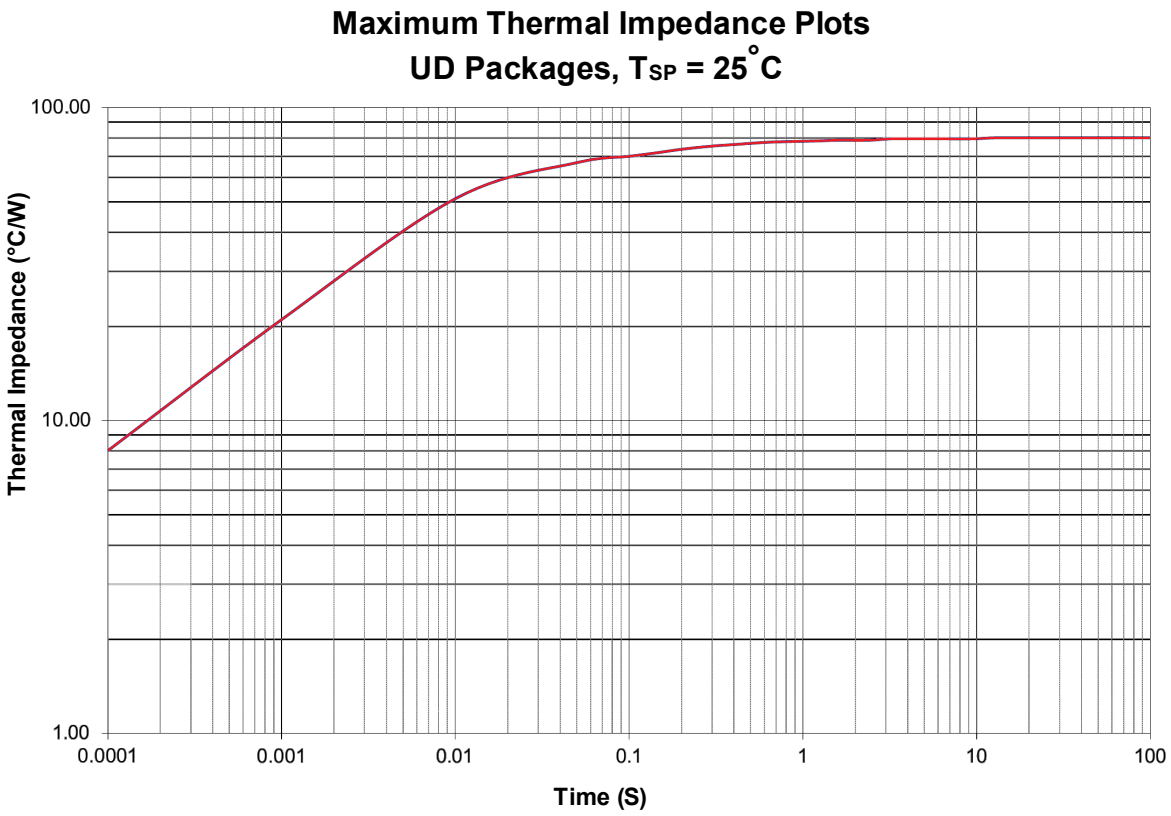


$R_{\theta JSP}$ = 120°C/W

NOTE: $Z_{\theta JX}$ = 75°C/W maximum at t_H = 10ms.

FIGURE 15. Thermal impedance for UB surface mount packages.

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NOTE: $Z_{\theta JX}$ = 51°C/W maximum at t_H = 10ms

FIGURE 16. Thermal impedance for UD surface mount packages.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Lead finish (see 3.4.1).

c. Packaging requirements (see 5.1).

d. Product assurance level and type designator.

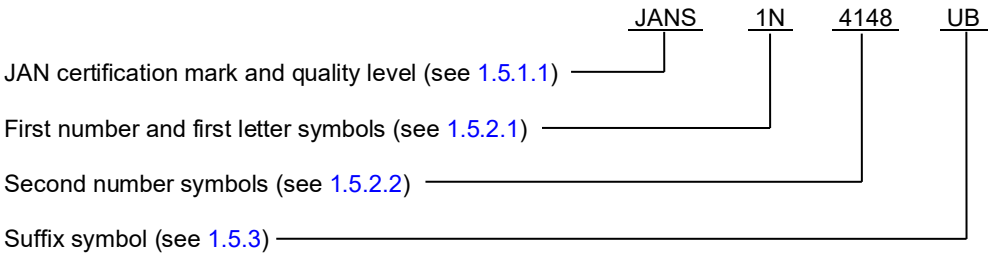
e. Destructive physical analysis when requested.

f. The complete PIN, see 1.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

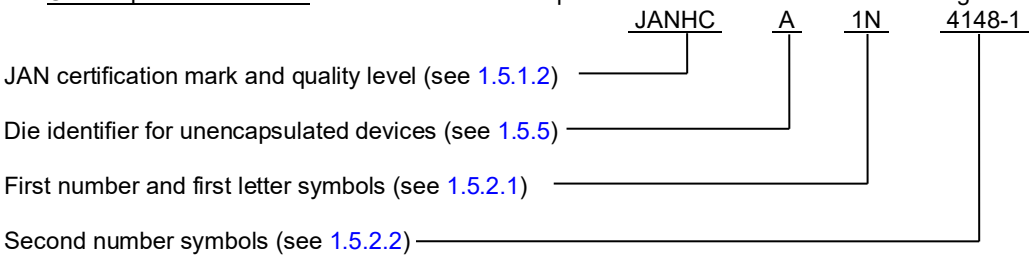
6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

6.5.1 List of encapsulated device types. PINs for type 1N914, 1N4148, and 1N4531.

JAN1N914	JANTX1N914	JANTXV1N914
JAN1N914UR	JANTX1N914UR	JANTXV1N914UR
JAN1N914UD	JANTX1N914UD	JANTXV1N914UD
JAN1N4148-1	JANTX1N4148-1	JANTXV1N4148-1
JAN1N4148UR-1	JANTX1N4148UR-1	JANTXV1N4148UR-1
JAN1N4148UB	JANTX1N4148UB	JANTXV1N4148UB
JAN1N4148UBCA	JANTX1N4148UBCA	JANTXV1N4148UBCA
JAN1N4148UBCC	JANTX1N4148UBCC	JANTXV1N4148UBCC
JAN1N4148UBCCC	JANTX1N4148UBCCC	JANTXV1N4148UBCCC
JAN1N4148UBD	JANTX1N4148UBD	JANTXV1N4148UBD
JAN1N4148UBCD	JANTX1N4148UBCD	JANTXV1N4148UBCD
JAN1N4148UB2	JANTX1N4148UB2	JANTXV1N4148UB2
JAN1N4148UB2R	JANTX1N4148UB2R	JANTXV1N4148UB2R
JAN1N4148UD	JANTX1N4148UD	JANTXV1N4148UD
JAN1N4531	JANTX1N4531	JANTXV1N4531
JAN1N4531UR	JANTX1N4531UR	JANTXV1N4531UR
JAN1N4531UD	JANTX1N4531UD	JANTXV1N4531UD

6.5.2 List of unencapsulated device types.

JANHCA1N4148	JANHCB1N4148	JANHCC1N4148	JANHCD1N4148	JANHCE1N4148
JANKCA1N4148	JANKCB1N4148	JANKCC1N4148	JANKCD1N4148	JANKCE1N4148
JANHCA1N914	JANHCB1N914	JANHCC1N914	JANHCD1N914	JANHCE1N914
JANKCA1N914	JANKCB1N914	JANKCC1N914	JANKCD1N914	JANKCE1N914
JANHCA1N4531	JANHCB1N4531	JANHCC1N4531	JANHCD1N4531	JANHCE1N4531
JANKCA1N4531	JANKCB1N4531	JANKCC1N4531	JANKCD1N4531	JANKCE1N4531

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6.6 Supersession information.

6.6.1 Discussion. Legacy design and construction of round body glass JANS1N4148 devices (with no suffix) were found to be unsuitable for space flight applications and have been deleted from this specification sheet. The replacement design, device type JANS1N4148-1 were also found to be unsuitable for space flight applications and is no longer in production, qualified, and prohibited for use. Device type JANS1N4148-1 in stock may be used provided the date code does not exceed 9208. Device type 1N4148 not packaged in a round glass body package are acceptable for space flight applications.

6.6.2 Obsoleted PINs. Device type JANS1N4148 (with no suffix) has been deleted from this specification sheet. Device types JANS1N4148-1 and 1N4148UR-1 are inactive for new design.

6.6.3 Cross reference substitution list. A PIN for PIN replacement table follows, and these devices are directly interchangeable.

Replacement PIN as specified within MIL-PRF-19500/578C, dated 17 January 1998	Superseded PIN as specified within MIL-S-19500/578, dated 1 March 1989	Replacement PIN as specified within MIL-S-19500/116J, dated 20 October 1992	Superseded PIN as specified within MIL-S-19500/116D, dated 22 June 1966
JANS1N6642	JANS1N6638	1N4148-1	1N4148
JANS1N6642	JANS1N6642	1N4148-1	1N4148
JANS1N6642US	JANS1N6638U	1N4148UR-1	
JANS1N6642US	JANS1N6642U	1N4148UR-1	

6.7 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N4148) will be identified on the QML.

JANHC and JANKC ordering information					
PIN	Manufacturer	Manufacturer	Manufacturer	Manufacturer	Manufacturer
	43611	52GC4	13409	13409	52GC4
1N4148	JANHCA1N4148, JANKCA1N4148	JANHCB1N4148, JANKCB1N4148	JANHCC1N4148, JANKCC1N4148	JANHCD1N4148, JANKCD1N4148	JANHCE1N4148, JANKCE1N4148

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6.8 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army − CR
Navy − SH
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, MI, SM
Navy − AS, MC
Air Force − 19

Preparing activity:
DLA − CC

(Project 5961−2025−031)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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